Computational Data Analysis (02582)

Clustering

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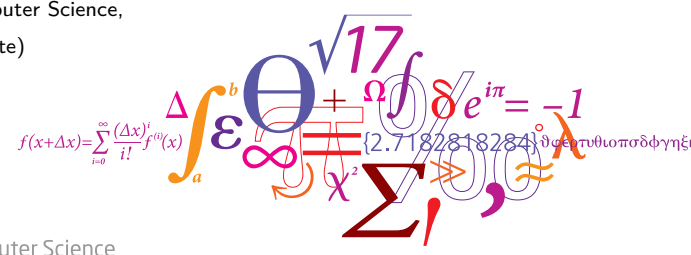
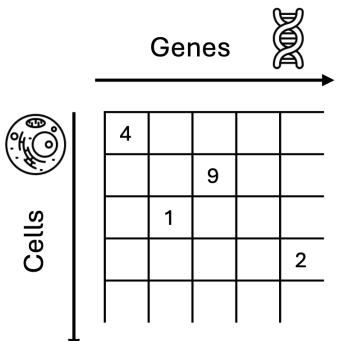
(Slides adapted from Line Clemmensen)

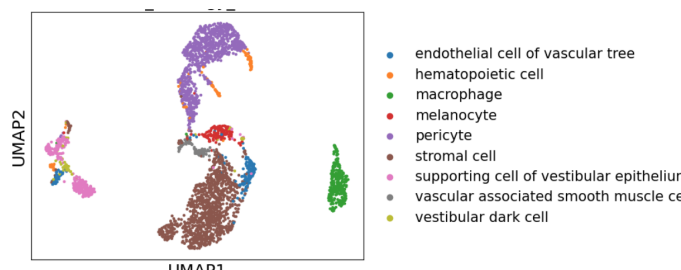


My Research

PhD in statistics (Mixed Effects Models for Large data sets)
Postdoc in Machine Learning for biological data

Single-cell RNA-sequencing data





(a) Tabula Sapiens [1] Ear data

[1] Tabula Sapiens Consortium, & Quake, S. R. (2025). Tabula Sapiens reveals transcription factor expression, senescence effects, and sex-specific features in cell types from 28 human organs and tissues. bioRxiv.

Agenda

- What is cluster analysis
- Similarity and dissimilarity measures
- K-means clustering
- Hierarchical clustering
- Gaussian Mixture Models (GMM)
- Validation and model selection
- Other clustering methods

Cluster Analysis

Unsupervised classification

- Separating or clustering observations
- Purpose: Seeing structure in data (gaining understanding), dimensionality reduction, outlier detection etc.
- Intuitive but vague definition

  Given an underlying set of points, partition them into a collection of **clusters** so that points in the same cluster are close together, while points in different clusters are far apart.

Similarity and dissimilarity measure

In what sense are points close in one cluster and far from points in another cluster?
How do we measure that?

Similarity and dissimilarity measure

In what sense are points close in one cluster and far from points in another cluster?
How do we measure that?

Similarity takes a large value when points are close.
Dissimilarity takes a large value when points are far apart. This reflects the **distance** between observations.
Any monotone-decreasing function can convert similarities to dissimilarities.
Both similarity and dissimilarity measures can be subjective. For example comparing the taste of three ice creams.

Clustering as an optimization problem

Defining the *variability* of points within a cluster:

$$Variability(c) = \sum_{i \in c} dist(mean(c), x_i)^2$$

Defining the dissimilarity of our clustering:

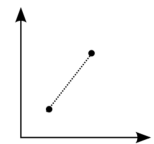
$$Dissimilarity(C) = \sum_{c \in C} variability(c)$$

Clustering as an optimization problem II

- What happens if we minimize dissimilarity?

Euclidean distance

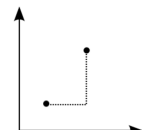
$$d(x_i, x_j) = \sqrt{\sum_{k=1}^{p} (x_{ik} - x_{jk})^2}$$



Useful for **quantitative** variables
Ordinal variables can be transformed to a quantitative scale.

Manhattan distance

$$d(x_i, x_j) = \sum_{k=1}^{p} |x_{ik} - x_{jk}|$$



Quantitative variables

Manhattan distance also called **city block distance** og **Hamming** distance.

Mahalanobis distance

$$d(x_i, x_j) = \sqrt{(x_i - x_j)^T \Sigma^{-1} (x_i - x_j)}$$

The distance is based on data itself and the two points are assumed to be of the same distribution with equal dispersion Σ.
Quantitative variables

Tanimoto distance (comparing sets, binary vectors)

$$d(x_i, x_j) = \frac{x_i^T x_j}{x_i^T x_i + x_j^T x_j - x_i^T x_j}$$

Let the sample x have $x_k = 1$ if it possesses the ith attribute, and $x_k = 0$ otherwise.
The ratio of the number of shared attributes to the number possessed by x_i or x_j.
Often used in information retrieval and biological taxonomy - works well for **categorical** variables.

Weighted distances

$$d(x_i, x_j) = \sum_{k=1}^{p} w_k d_k(x_{ik}, x_{jk}),$$

$$\sum_{k=1}^{p} w_k = 1$$

Give different weight to the p attributes (variables).
w_k indicates that attribute k uses an individual dissimilarity and all p dissimilarities are then combined into one total dissimilarity (d) for all attributes.
Note that setting $w_k = 1/p$ does *not* necessarily give equal influence to the attributes.
We would have to normalize with the average distance for the kth attribute.

Unsupervised clustering

We have data but no information about class belonging

- Group data in clusters
- Observations that are "near" each other should belong to the same class/cluster
- This can help us unveil an unknown structure in data
- It is like classification but without an answer, ie unsupervised

K-means clustering

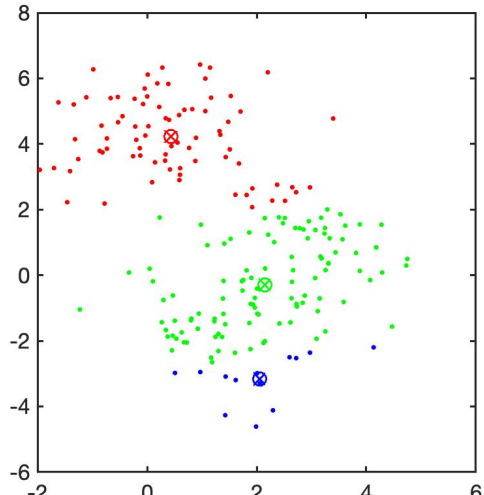
- Super simple algorithm for clustering

K-means clustering

Decide how many clusters, K, there should be, and randomly initialize K cluster centers (centroids).
Alternating between two steps (until assignments do not change)

- Assign each point to the closest cluster center (centroid)
- Compute new cluster centers (centroids) according to the assignments

Example - initialize 3 centroids in dense areas



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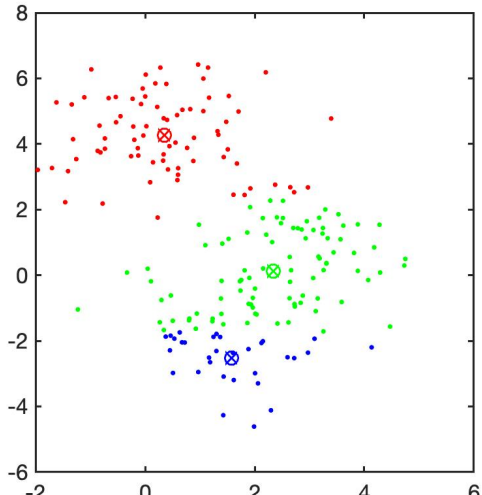

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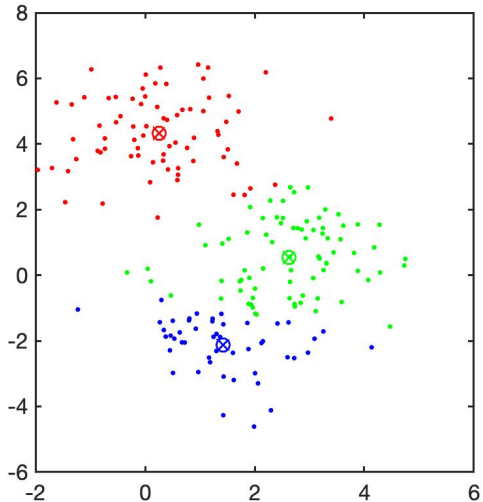

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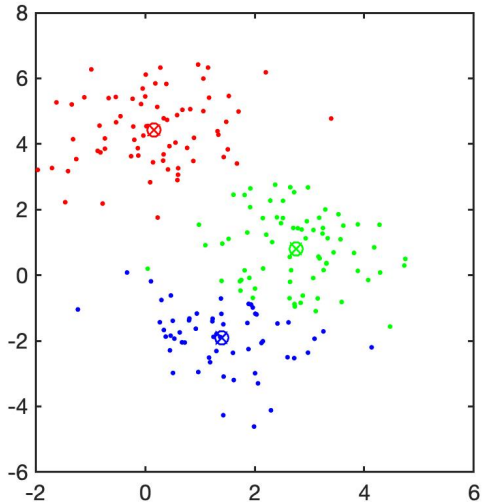

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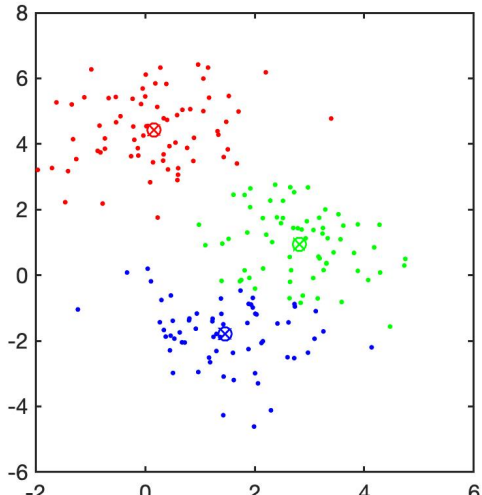

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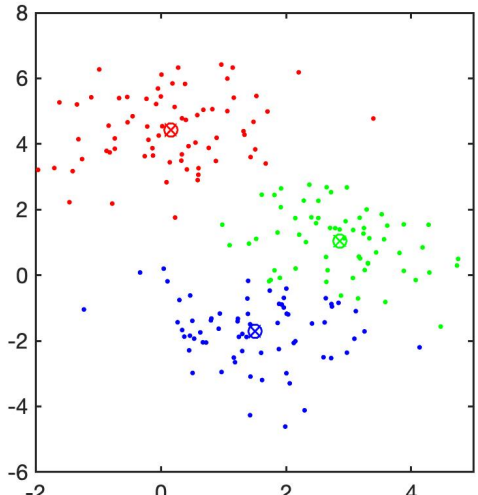

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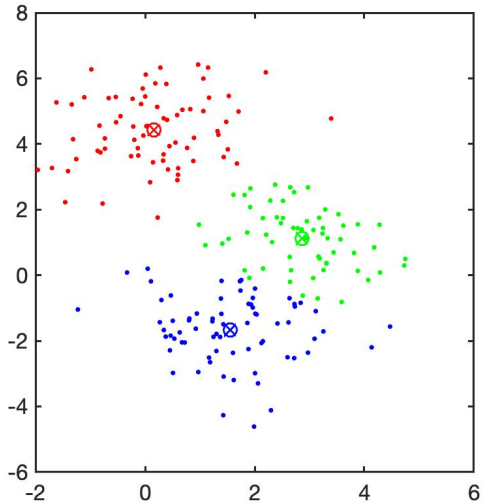

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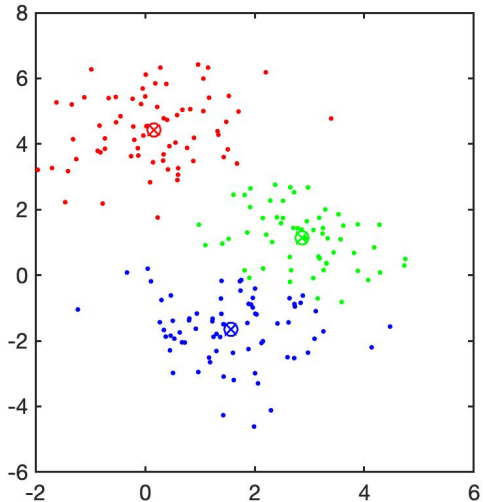

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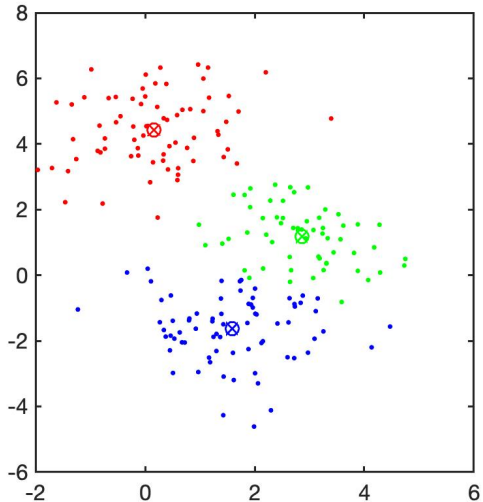

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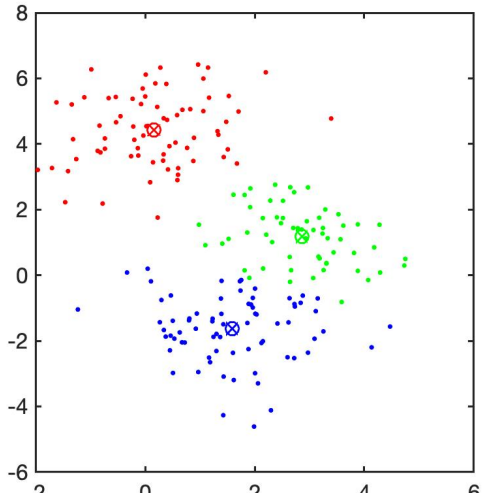
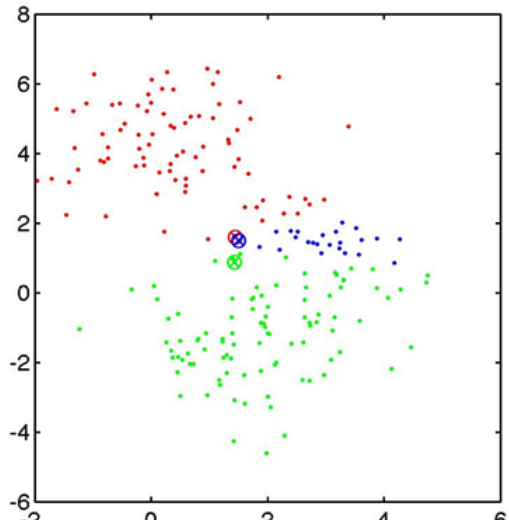


Converged

Example - initialize 3 centroids close to common center of gravity



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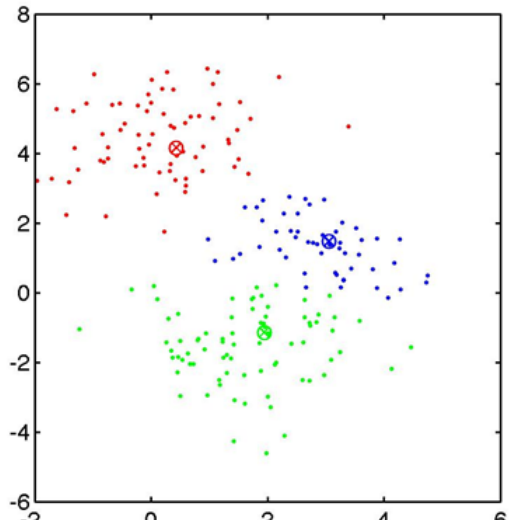

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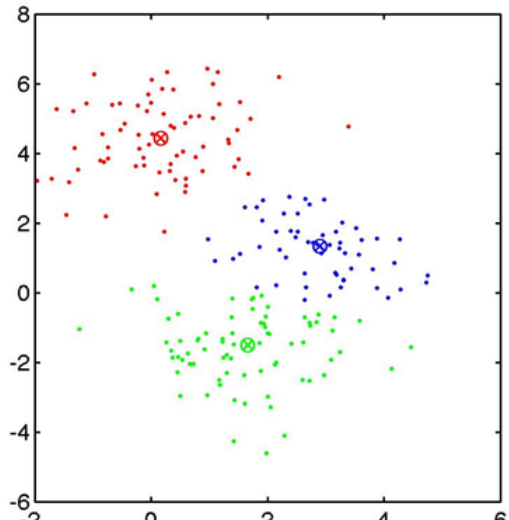

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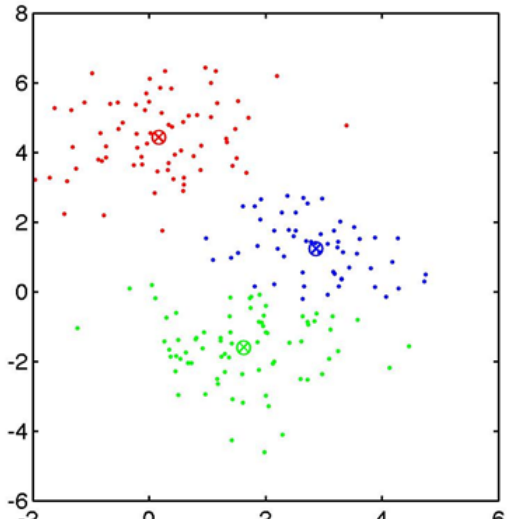
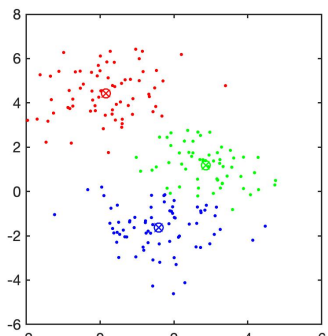


Converged

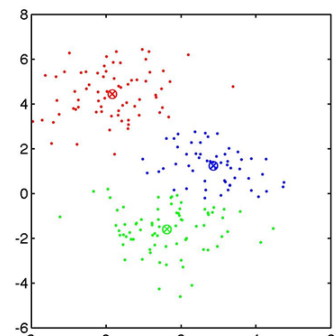
K-means

- Solution may change depending on initialization (randomness of the solution)
- Run-time (number of iterations to convergence) is dependent on initialization

Dense area, 10 iterations



Center of gravity, 4 iterations



K-medoids

- K-medoids use one of the observations as cluster center
    - Computationally much heavier than K-means
- Makes K-medoids more robust to outliers than K-means
- Also refered to as PAM, Partitioning Around Medoids

Hierarchical Clustering

- Generates a tree of observations (dendrogram)
- Each level of the tree reflects one number of clusters
- From all data in one cluster down to one observation in each cluster

Hierarchical clustering

- Two approaches
  - Bottom-up: Agglomerative (commonly used)
    Start with one observation in each cluster and combine them to form bigger clusters
  - Top-down: Divisive
    Start with all observations in one cluster and divide them to form smaller clusters
- A full tree gives $n - 1$ levels in the hierarchy
- At each level perform the split or the merge which gives the largest or smallest between-group dissimilarity, depending on which approach you take
- Do not require input of number of clusters
- Uses dissimilarity between clusters

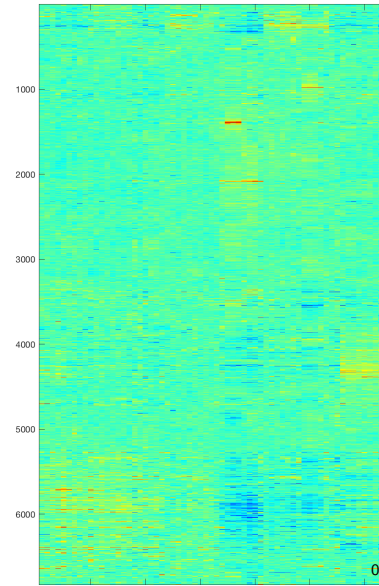
Microarray example

- The mRNA microarray data from ESL, example 4
- Samples from cancer tumors
- 6830 genes
- 64 samples

Unsupervised clustering:

Which samples are most similar to each other, in terms of their expression profiles across genes.

Do similar samples share the same form of cancer?



Cluster-cluster distance

Complete-linkage hierarchical clustering

This cluster-cluster distance is measured as the distance from the furthest pair of points (i, j) from the clusters (G, H) respectively.

$$d_{CL} = \max_{\substack{i \in G \\ j \in H}} d_{ij}$$

Distance between observations

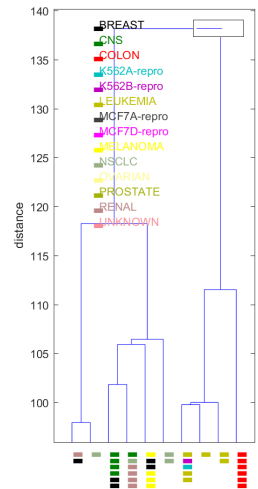
Distance between clusters

Also called the **furthest-neighbor** technique

Microarray example - complete-linkage clustering

- Approach: Top-down (divisive)
- Sample-sample: Euclidean
- Cluster-cluster: Complete
- Dendrogram cut at 10 nodes

Tends to give **balanced trees** independent of data



A **dendrogram** consists of many U-shaped lines connecting objects in a hierarchical tree.

The **height** of each U represents the distance between the two objects being connected.

Cluster-cluster distance

Single-linkage hierarchical clustering
Cluster-cluster distance measured as the distance of the **closest** pair of points (i, j) from the clusters (G, H) respectively.

$$d_{SL} = \min_{\substack{i \in G \\ j \in H}} d_{ij}$$

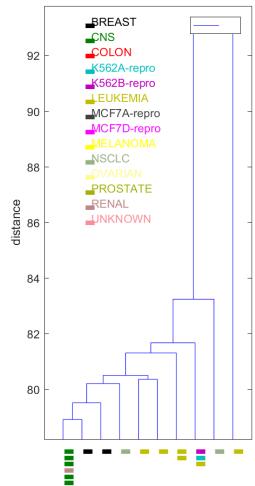
Also called the **nearest-neighbor** technique

Chaining can be a problem

Microarray example - single-linkage clustering

- Approach: Top-down (divisive)
- Sample-sample: Euclidean
- Cluster-cluster: Single-linkage
- Dendrogram cut at 10 nodes

Tends to give **unequal cluster sizes** independent of data



Cluster-cluster distance

Ward-linkage hierarchical clustering

Cluster-cluster distance measured as the increment in within-cluster sum of squares, when merging clusters G and H

$$d_{Ward} = \sqrt{n_G n_H \frac{||\bar{x}_G - \bar{x}_H||_2^2}{(n_G + n_H)}}$$

Ward's distance measures how much the sum of squares will increase when we merge the two clusters,

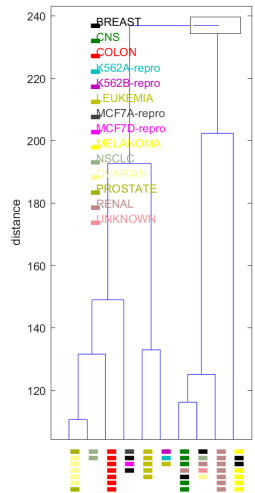
$$d_{Ward} = \sum_{i \in G \cup H} ||x_i - \bar{x}_{G \cup H}||_2^2 - \sum_{i \in G} ||x_i - \bar{x}_G||_2^2 - \sum_{i \in H} ||x_i - \bar{x}_H||_2^2$$

Microarray example - Ward-linkage clustering

- Approach: Top-down (divisive)
- Sample-sample: Euclidean
- Cluster-cluster: Ward-linkage
- Dendrogram cut at 10 nodes

Tends to give a good **compromise** between balanced/unbalanced clusters

Can only be used with Euclidean distance



Bottom-up example

	Copenhagen	Nuuq	Edinburgh	Reykavik
Copenhagen	0 km	3535 km	983 km	2107 km
Nuuq	3535 km	0 km	2765 km	1430 km
Edinburgh	983 km	2765 km	0 km	1374 km
Reykavik	2107 km	1403 km	1374 km	0 km

{Copenhagen} {Nuuq} {Edinburgh} {Reykavik}

Bottom-up example

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{Copenhagen} {Nuuq} {Edinburgh} {Reykavik}
{Copenhagen, Edinburgh} {Nuuq} {Reykavik}

Which clusters should we merge if we use **single linkage** ?
Which clusters should we merge if we use **complete linkage** ?

Bottom-up example

	Copenhagen	Nuuq	Edinburgh	Reykavik
Copenhagen	0 km	3535 km	983 km	2107 km
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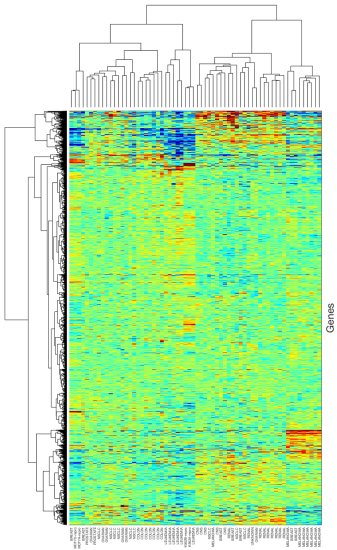
{Copenhagen, Edinburg, **Reykavik** } {Nuuq} *single linkage*
or
{Copenhagen, Edinburg} {Nuuq, **Rekavik** } *complete linkage*

Two way clustering (biclustering)

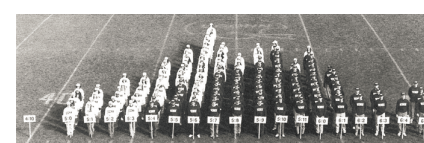


- Cluster both genes and samples
- Presented as a heatmap

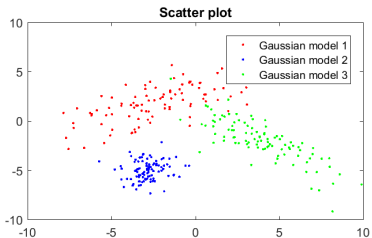
In Python: `SpectralBiclustering` function in `scikitlearn`

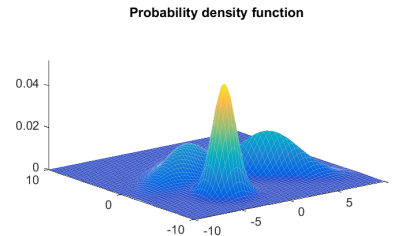
Gaussian Mixture Models (GMM) and Expectation-Maximization Algorithm (EM algorithm)

- Probabilistic model giving *soft* cluster assignments
- Data belong to one of several Gaussian distributions
- An un-observed (latent) random variable selects which distribution the observation comes from.
- This gives a (complicated) likelihood function
- Easily solved using the EM algorithm



Gaussian models





Observations from three Gaussian distributions

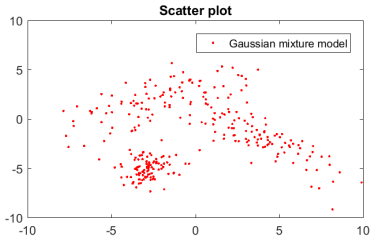
$$\begin{cases} X_i \in N(\mu_1, \Sigma_1) & \text{if } Z_i = 1 \\ X_i \in N(\mu_2, \Sigma_2) & \text{if } Z_i = 2 \\ X_i \in N(\mu_3, \Sigma_3) & \text{if } Z_i = 3 \end{cases}$$

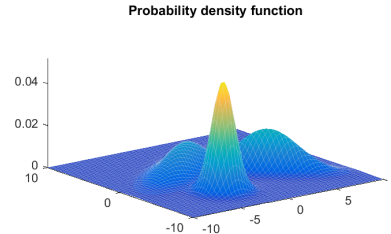
Known variables, Z_i, indicate cluster/distribution

$$\begin{cases} Z_i = 1 & i = 1, ..., 100 \\ Z_i = 2 & i = 101, ..., 200 \\ Z_i = 3 & i = 201, ..., 300 \end{cases}$$

What are the unknown model parameters?

Gaussian mixture models





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Un-observed variables, Z_i, indicate cluster/distribution

$$\begin{cases} P(Z_i = 1) = \pi_1 \\ P(Z_i = 2) = \pi_2 \\ P(Z_i = 3) = \pi_3 \end{cases}$$

The probabilities sum to one,

$$\pi_1 + \pi_2 + \pi_3 = 1$$

What are the unknown model parameters?

Likelihood for the Gaussian mixture example

Parameters in the Gaussian mixture model are,

$$\boldsymbol{\theta} = (\pi_1, \pi_2, \pi_3, \mu_1, \mu_2, \mu_3, \Sigma_1, \Sigma_2, \Sigma_3)$$

and

$$\mathbf{Z} = (Z_1, ..., Z_n) \qquad (\ n \text{ might be very large})$$

The joint log-likelihood of the observations and the latent variables is

$$\ell(\boldsymbol{\theta}; \mathbf{x}, \mathbf{Z}) = \log p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{Z}) = \sum_{i=1}^{n} \sum_{j=1}^{3} \mathbb{I}_{\{Z_i = j\}} \left(\log N(x_i; \mu_j, \Sigma_j) + \log \pi_j \right)$$

with maximum likelihood estimate

$$\boldsymbol{\theta}_{ML} = \arg\max_{\boldsymbol{\theta}, \mathbf{Z}} \ell(\boldsymbol{\theta}; \mathbf{x}, \mathbf{Z})$$

Finding a solution $(\boldsymbol{\theta}, \mathbf{Z})$ is much simplified by the **EM-algorithm**.

The Gaussian mixture EM-algorithm I

1. Initialize means μ_j, covariances Σ_j, and probabilities π_j.

2. **Expectation step**
   Compute the posterior probabilities $\gamma_{ij} = p_{\boldsymbol{\theta}^{(k)}}(Z_i = j|x_i)$, where $\boldsymbol{\theta}^{(k)}$ is the current estimate of the parameters. We can do this by Bayes formula

$$\gamma_{ij} = p_{\boldsymbol{\theta}^{(k)}}(Z_i = j|x_i) = \frac{p_{\boldsymbol{\theta}^{(k)}}(x_i|Z_i = j)p_{\boldsymbol{\theta}^{(k)}}(Z_i = j)}{p_{\boldsymbol{\theta}^{(k)}}(x_i)} = \frac{\pi_j^{(k)} N(x_i; \mu_j^{(k)}, \Sigma_j^{(k)})}{\sum_{j=1}^{3} \pi_j^{(k)} N(x_i; \mu_j^{(k)}, \Sigma_j^{(k)})}$$

The Gaussian mixture EM-algorithm II

This step is based on derivation of the expectation of the log-likelihood with respect to the conditional distribution of $\mathbf{Z}$ given $\mathbf{x}$ and $\boldsymbol{\theta}^{(k)}$, denoted $Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)})$ such that

$$
\begin{aligned}
Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)}) = \mathbb{E}_{\mathbf{Z}|\mathbf{x},\boldsymbol{\theta}^{(k)}}\left[\log p_{\boldsymbol{\theta}}(\mathbf{x},\mathbf{Z})\right] &= \mathbb{E}_{\mathbf{Z}|\mathbf{x},\boldsymbol{\theta}^{(k)}}\left[\sum_{i=1}^{n}\sum_{j=1}^{3}\mathbb{I}_{\{Z_i=j\}}\left(\log N(x_i;\mu_j,\Sigma_j)+\log\pi_j\right)\right] \\
&= \sum_{i=1}^{n}\sum_{j=1}^{3}\mathbb{E}_{\mathbf{Z}|\mathbf{x},\boldsymbol{\theta}^{(k)}}\left[\mathbb{I}_{\{Z_i=j\}}\right]\left(\log N(x_i;\mu_j,\Sigma_j)+\log\pi_j\right) \\
&= \sum_{i=1}^{n}\sum_{j=1}^{3}p_{\boldsymbol{\theta}^{(k)}}(Z_i=j|x_i)\left(\log N(x_i;\mu_j,\Sigma_j)+\log\pi_j\right)
\end{aligned}
$$

The Gaussian mixture EM-algorithm III

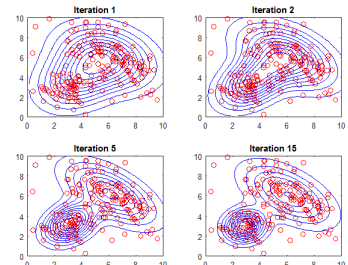
3. **Maximization step**
   Calculate the new estimates of the parameters using γ_{ij} such that

$$\mu_j^{(k+1)} = \frac{\sum_{i=1}^n \gamma_{ij} x_i}{\sum_{i=1}^n \gamma_{ij}} \quad \Sigma_j^{(k+1)} = \frac{\sum_{i=1}^n \gamma_{ij}(x_i - \mu_j)(x_i - \mu_j)^\top}{\sum_{i=1}^n \gamma_{ij}} \quad \pi_j^{(k+1)} = \frac{1}{n}\sum_{i=1}^n \gamma_{ij}$$

   These formulas are solutions to the maximization of the quantity $Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)})$.

4. Iterate until convergence

Gaussian mixture simulation



Simulation of a Gaussian Mixture model

$$P(Z_i = 1) = 0.3$$
$$P(Z_i = 2) = 0.7$$
$$X_i \in N\left(\begin{bmatrix} 3 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}\right) \quad \text{if } Z_i = 1$$
$$X_i \in N\left(\begin{bmatrix} 6 \\ 6 \end{bmatrix}, \begin{bmatrix} 4 & -2 \\ -2 & 4 \end{bmatrix}\right) \quad \text{if } Z_i = 2$$

and $i = 1 \ldots 200$

Some tricks

- High dimension (large p) and few data (small n)
  - Postulate same covariance structure for all clusters, $\Sigma = \Sigma_1 = ... = \Sigma_K$
  - Postulate diagonal covariance structure, $\Sigma = \mathrm{diag}(\sigma_1, ..., \sigma_K)$
  - Regularize, $\Sigma = \Sigma + \lambda I$
  - Use first principal components instead of all dimensions in data

The problem of validation

- How to select the number of clusters?

Selecting the number of clusters

- Silhouette method (heuristic)
- Gap-statistics
- Goodness-of-fit measures (when a distribution is assumed (for example GMM))
    - Chi-squared statistics
    - Kolmogorov-Smirnov statistics
    - AIC and BIC
- Biological or physical interpretation

Why not cross validation?

Silhouette method - heuristic

Let $a(i)$ denote the average distance between observation i and all other observations assigned to the same cluster.
Let $b(i)$ denote the average distance between observation i and all observations assigned to the neighboring cluster (cluster where i is not a member and where average distance is largest).
Silhouette is defined as

$$s(i) = \frac{b(i) - a(i)}{max\{a(i), b(i)\}}$$

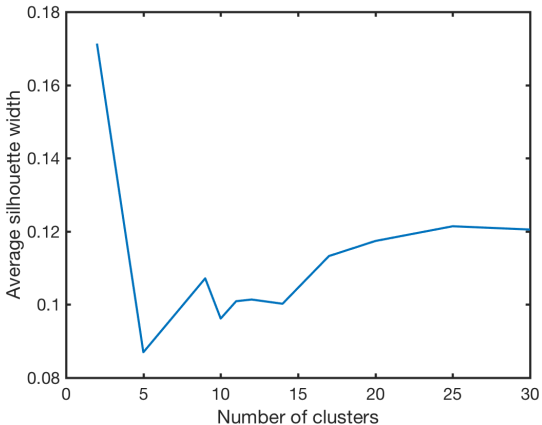
The average $s(i)$ over all observations is an estimate of how appropriately the points are clustered. Select number of clusters when all clusters have observations above average silhouette width, or choose K^* with the maximum average silhouette.

- Favour convex and spherical clusters
- Outlier sensitive

Silhouettes: a graphical aid to the interpretation and validation of cluster analysis, Rousseeuw (1987)
Shortcomings of silhouette in single-cell integration benchmarking, Ohler (2025)

Example - Zip data

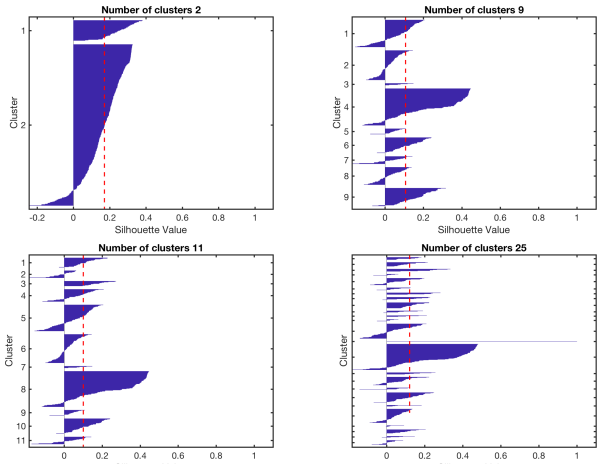
Select clusters based on the maximum average silhouette score.



Optimal $K^* = 2$ or 25?

Example - Zip data

Select the number of clusters when all clusters have observations above average silhouette score



Using silhouette plots: Optimal $K^* = 2$ or 9?

Within cluster dissimilarity

Distance between all points in one cluster (Euclidean distance)

$$
\begin{aligned}
D_\ell &= \sum_{\substack{C_k(i)=\ell \\ C_k(j)=\ell}} ||x_i - x_j||^2 \\
&= \ldots \\
&= N_\ell \sum_{C_k(i)=\ell} ||x_i - \bar{x}_\ell||^2
\end{aligned}
$$

Within cluster dissimilarity

$$
W_k = \sum_\ell \frac{1}{2N_\ell} D_\ell
$$

($C_k(i) =$ cluster for obs. i when we have k clusters in the model.)

Tibshirani, Walter, & Hastie; Estimating the number of clusters... 2001, Eq 1 and 2

Gap-statistic

Compares the log criterion value with K clusters to the expected log criterion value for **uniformly distributed** data (20 simulations)

$$G(K) = log(U_k) - log(W_k)$$

- U_k within cluster dissimilarity, simulated data - mean over 20 samples
- W_k within cluster dissimilarity, actual data

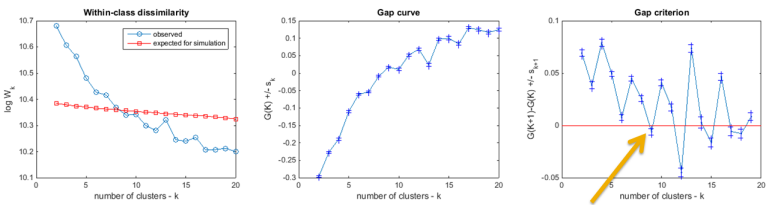
Choose

$$K^* = \arg\min_k \{K | G(K) \geq G(K+1) - s'_{K+1}\}$$

where

$$s'_{K+1} = \mathrm{std}(\log(U_K))\sqrt{1 + 1/20}$$

Example - zip data



Optimal $K^* = 9$

Based on simulation - might differ from one simulation to another

Gap-statistic

- Can be used both for K-means, K-medoids and hierarchical clustering
- Works with different measures of the within-cluster dissimilarity

A warning...

Clustering will **always** generate a grouping

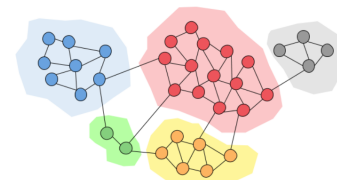
- Even when data are completely random
- Be careful with the interpretation
- It is the application that tells if the structure is relevant (use your **domain knowledge**)

Other clustering methods

- Louvain/Leiden clustering (network based)

  Standard pipeline in scRNA-seq data:
  PCA + Louvain/Leiden clustering

- HDBSCAN (density based)
- Deep clustering (neural networks)
- Even more exists... (see the literature)



Medium: *Leiden Clustering for Community Detection: A Step-by-Step Guide with Python Implementation*

Exercise - K-means clustering

1. Run K-means clustering using kmeans_demo (Kmeans_demo.m, kmeans_demo.R, or kmean_demo.py). The demo fits K-means clustering to tree classes each sampled from a 2D mixture of Gaussian distributions, you can change the values of K in the code.
   - Try different numbers of clusters. Which value(s) of K seem reasonable? Why?

Exercise - Hierarchical clustering

2. Perform hierarchical clustering on the zip data (example 3 in ESL). The data consist of 400 samples of handwritten digits 0-9 in 16×16 grayscale images (= 256 features).
   - Try different dissimilarity measures. Which work best?
   - Where would you cut the dendrogram (ie how many clusters should we choose)?
   - You may use hierarchicalEx.m, HierarchicalEx.py, or hierarchicalEx.R.

Exercise - Gap statistic

3. Use the gap-statistic to select K for K-means clustering on the zip data.Use the gap-statistic to select K for K-means clustering on the zip data.
   - You will need to write the calculations of the within-class dissimilarities and the gap statistic yourself.
   - Use KmeansEx.m, KmeansEx.R, or kMeans_ex.py
   - You may use hierarchicalEx.m, HierarchicalEx.py, or hierarchicalEx.R.
   - Try first kmeans and then kmedoids (Optional).

Exercise - Gaussian mixture

4. We have data with four different measures from flowers of three different species (*Fisheriris.csv*). There are 50 observations of each species. See if you can identify three clusters in data using gaussian mixture modelling. (Two of the species are very similar)

Iris setosa

Iris versiocolor

Iris verginica

- Plot data using a scatterplot matrix
- Loop over different numbers of clusters
- Plot BIC/AIC for different model orders
- Notice the different extra parameters in the provided Gaussian Mixture function - they might be necessary.

COURSE OVERVIEW*

Week	Date	Topics	Lecturer	Literature
1	06/2	Introduction (EPE, bias, variance, collinearity, OLS, Ridge, KNN, K-means)	Sneha	ESL: 1, 2.1-2.3, 2.9, 3.1, 3.2, 3.4.1, 4.1 & 13.3
2	13/2	Model selection (CV, Bootstrp, Cp, AIC, BIC);	Sneha	ESL: chapter 7. You may safely skip sections 7.8 & 7.9
	Week 8	**Case 1 handout**		
3	20/2	Sparse regression and multiple testing (LASSO, Elastic net, Bonferroni, BH, FDR)	Line	ESL: 3.3, 3.4, 3.6, 18.1, 18.7
4	27/2	Linear classifiers (LDA, QDA, Log reg, splines, ROC)	Sneha	ESL: 4.3, 4.4, 5.1 & 5.2
5	06/3	Tree based methods (CART and bagging)	Mathies	ESL: 9.2, 8.7
6	13/3	Ensemble methods (Boosting and random forests)	Sneha	ESL: chapters 10 & 15
7	20/3	Basis expansions, Support Vector Machines and convex optimization	Line	ESL: 4.5, 12.1, 12.2 & 12.3.1
8	27/3	Subspace methods (PCA, SPCA, CCA, PLS)	Sneha	ESL: 14.5.1, 14.5.5, 3.5, 3.7
	30/3	**Deadline Case 1 (Short report) 23:59**		
9	03/4	Unsupervised clustering (Hierarchical clustering, K-means, Gaussian Mixtures, Gap statistics). Case 2 handout	Sneha	ESL: 14.3
10	10/4	ANN & SOM	Sebastian	ESL: 11.1-11.5 & 14.4
11	24/4	Case 1 discussion and case competition. Unsupervised decompositions (SC, NMF, AA, ICA)	Sneha	ESL: 14.6, 14.7, and article 'sparse coding'
12	01/5	Multiway models	Mathies	WireOverview.pdf
13	**08/5**	**Case 2 poster peer review 11:45 + TBD**		**TBD**
	19/5	**Written Examination**		

***Subject to minor changes**